# Shenglai Zeng

shenglaizeng@gmail.com | (+1) 517-974-7616 | Website | LinkedIn | Google Scholar | Citations 1340+

## Research Interest

I am a fourth-year Ph.D. candidate at Michigan State University, advised by Professor Jiliang Tang, and I am on the job market for positions starting in 2027. My research builds **personalized, proactive, and safe agents** with **multimodal capabilities** that interact with people. I work on three fronts: the **agent harness** (memory, retrieval, and self-improvement) that determines what an agent can actually do, **post-training** that makes it better, and **evaluation and environments** that reveal where agents fail. I have published 35+ papers at NeurIPS, ICML, ICLR, ACL, EMNLP, NAACL, KDD, and CVPR, received the Best Paper Award of IEEE Transactions on Cloud Computing (2023) and an ACL-2026 SAC Highlight (top 1%, Best Paper Candidate), and have industry research experience at Meta AI, Amazon, and Baidu.

## Education

**Michigan State University, East Lansing, USA** Sept. 2023 – Present
*Ph.D. Candidate in Computer Science and Engineering*

- Advisor: Dr. Jiliang Tang | Lab: Data Science and Engineering (DSE) Lab
- Expected Graduation: **May 2027**
- Research Focus: Agentic AI, Retrieval & Memory, Personalized LLMs, Agent Safety & Privacy, Multimodal

**University of Electronic Science and Technology of China, Chengdu, China** Sept. 2019 – Jun. 2023
*B.Sc. in Computer Science and Engineering, Yingcai Honor School*

- CGPA 3.98/4.00; Weighted Average 93.97/100 (**1st among 100 students**)

## Honors and Awards

- **SAC Highlight (Top 1%) & Best Paper Candidate**, ACL-2026 2026
- **Best Paper Award**, IEEE Transactions on Cloud Computing (JCR Q1) 2023
- **The Most Outstanding Students Award**, UESTC (highest honor; 10 students university-wide)
- **National Scholarship** (highest undergraduate honor in China) 2020
- WAC Scholarship (10 undergraduates university-wide per year), UESTC 2021
- 1st Outstanding Academic Scholarship (Top 5%), UESTC 2020, 2021, 2022

## Industry Experience

*All papers in this section are **first-authored** (including co-first).*

**Meta AI, Seattle, USA** Summer 2026
*Research Scientist Intern, Next Generation Proactive Agents*
Mentored by Xiao Yang, Luna Dong, and Scott Yih
Closely worked with a cross-organizational team spanning Superintelligence Labs, Reality Labs, FAIR, and MRS

I built **memory-driven proactive agents**: assistants that can proactively use their context and decide *whether and when* to speak, not just what to answer. I introduced **ProPerMem**, a benchmark for memory-driven proactive assistance (**MPA**) pairing egocentric video with year-long, distractor-rich personal memories (∼**618K tokens** each) and scoring content and timing jointly, and found **existing models severely limited** in this regime. I

then built **ProPerAgent**, whose plug-and-play harness surfaces memory from a graph-organized store and gates it down to what is worth interrupting for, lifting strict accuracy from **less than 10%** to **37.5%** (Qwen3.8-27B) and to **38%** on GPT-5.3-Chat with no training at all, reaching **42.4%** with **RL post-training**.

- ▷ ProPerAgent: Toward Memory-Driven Proactive Assistance **[Submitted to ICLR'27]**.

**Amazon, Search & Shopping Foundation Model Team, CA, USA** Summer 2024, 2025
*Applied Scientist Intern, Retrieval-Augmented Generation and Personalization*
Mentors: Tianqi Zheng, Dante Everaert | Managers: Hanqing Lu, Chuan Tian, Monica Xiao Cheng

My work centered on two sides of making retrieval-augmented and personalized LLMs deployable. **On the model side** (how a system uses the context it retrieves): attention-guided compression cuts personalization context **50**× (10K→200 tokens) at near-full-context quality [ACL'26]; gated representation fine-tuning buys **+26–35 points** of context robustness while tuning only **0.0004% of parameters on fewer than 200 examples** [ACL'25]; and representation-based knowledge checking filters misleading evidence at **95% accuracy** [NAACL'25]. **On the data side** (what the retrieval corpus itself exposes): **self-improving synthetic data** replaces private records at comparable task quality [EMNLP'25], and I gave the first account of how **multimodal RAG leaks** the images and text it retrieves [EMNLP'25]. Invited back for a second summer.

- ▷ Efficient Personalized LLMs (PLLMs) [ACL'26, Oral, Best Paper Candidate, SAC Highlight].
- ▷ Knowledge Checking in RAG [NAACL'25, Oral].
- ▷ Representation Finetuning for Robust RAG [ACL'25].
- ▷ Self-Improving Synthetic Data for RAG [EMNLP'25].
- ▷ Multi-modal RAG [EMNLP'25].

**Baidu Inc., Search & Agent Business Team, Beijing, China** May 2023 – May 2024
*Research Intern, RAG Privacy and LLM Memorization*
Mentors: Dr. Yiding Liu, Dr. Dawei Yin

I established the privacy foundations of retrieval-augmented generation: the **first work to reveal that retrieval-augmented systems carry severe privacy risk**, leaking their own retrieval database [ACL'24]. The threat model it introduced has since been carried into the privacy analysis of future systems (multi-modal RAG, agent memory, and multi-agent systems), and the paper now has **350+ citations**. I also produced the **first task-level map of memorization in LLM post-training** [ACL'24], showing how memorization **scales with model size** and that the scaling itself is task-dependent (22.3% for summarization vs. 0.15% for reading comprehension).

- ▷ RAG Privacy and Retrieval-Database Leakage [ACL'24].
- ▷ Memorization in LLM post-training [ACL'24].

## Publications

35+ papers at top-tier ML/NLP venues (NeurIPS, ICML, ICLR, ACL, EMNLP, NAACL, KDD, CVPR), including **12 first-authored** publications. **1340+ citations** (Google Scholar, Sept. 2026). Recognitions include a **Best Paper Award** (IEEE TCC) and **Best Paper Candidate** (ACL), **Spotlight** (ICLR), **Oral** presentations (ACL, NAACL), and adoption in frontier-model technical reports (Muse Spark, Qwen, Kimi). ** denotes equal contribution.*

### First-Authored Publications

**[NeurIPS'26]** **S. Zeng**, Q. Wang, K. Guo, X. Dai, X. Long, H. Liu, "Magnifying What Matters: Attention-Guided Adaptive Rendering for Visual Text Comprehension", *Conference on Neural Information Processing Systems*, 2026.

**[ICML'26]** **S. Zeng***, J. Zhang*, K. Guo, X. Dai, H. Liu, J. Tang, Y. Chang, "Fix Before Search: Benchmarking Agentic Query Visual Pre-processing in Multimodal Retrieval-Augmented Generation", *International Conference on Machine Learning*, 2026.

**[ACL'26, Oral]** **S. Zeng**, T. Zheng, C. Tian, D. Everaert, Y.-S. Wang, Y. Huang, M. J. Morais, R. Patki, J. Tian, X. Dai, K. Guo, M. X. Cheng, H. Liu, "Attn-GS: Attention-Guided Context Compression for Efficient Personalized LLMs", *Annual Meeting of the Association for Computational Linguistics,* ***Oral***, ***SAC Highlight (Top 1%)***, ***Best Paper Candidate***, 2026.

**[EMNLP'25]** **S. Zeng**, J. Zhang, P. He, J. Ren, T. Zheng, H. Lu, H. Xu, H. Liu, Y. Xing, J. Tang, "Mitigating the Privacy Issues in Retrieval-Augmented Generation (RAG) via Pure Synthetic Data", *Conference on Empirical Methods in Natural Language Processing*, 2025.

**[EMNLP'25]** **S. Zeng***, J. Zhang*, J. Ren, T. Zheng, H. Liu, X. Tang, Y. Chang, "Beyond Text: Unveiling Privacy Vulnerabilities in Multi-modal Retrieval-Augmented Generation", *Conference on Empirical Methods in Natural Language Processing*, 2025.

**[ACL'25]** **S. Zeng**, P. He, K. Guo, T. Zheng, H. Lu, Y. Xing, H. Liu, "Towards Context-Robust LLMs: A Gated Representation Fine-tuning Approach", *Annual Meeting of the Association for Computational Linguistics*, 2025.

**[NAACL'25, Oral]** **S. Zeng**, J. Zhang, B. Li, Y. Lin, T. Zheng, D. Everaert, H. Lu, H. Liu, Y. Xing, M. X. Cheng, J. Tang, "Towards Knowledge Checking in Retrieval-Augmented Generation: A Representation Perspective", *Conference of the North American Chapter of the Association for Computational Linguistics,* ***Oral***, 2025.

**[ACL'24]** **S. Zeng***, Y. Li*, J. Ren, Y. Liu, H. Xu, P. He, Y. Xing, S. Wang, J. Tang, D. Yin, "Exploring Memorization in Fine-tuned Language Models", *Annual Meeting of the Association for Computational Linguistics*, 2024.

**[ACL'24]** **S. Zeng***, J. Zhang*, P. He, Y. Xing, Y. Liu, H. Xu, J. Ren, S. Wang, D. Yin, Y. Chang, J. Tang, "The Good and The Bad: Exploring Privacy Issues in Retrieval-Augmented Generation (RAG)", *Findings of the Association for Computational Linguistics: ACL*, 2024.

**[IEEE TCC, Best Paper]** **S. Zeng**, Z. Li, H. Yu, Z. Zhang, L. Luo, B. Li, D. Niyato, "HFedMS: Heterogeneous Federated Learning with Memorable Data Semantics in Industrial Metaverse", *IEEE Transactions on Cloud Computing,* ***Best Paper Award***, 2023.

**[DASFAA'22, Oral]** **S. Zeng**, Z. Li, H. Yu, Y. He, Z. Xu, D. Niyato, H. Yu, "Heterogeneous Federated Learning via Grouped Sequential-to-Parallel Training", *International Conference on Database Systems for Advanced Applications,* ***Oral***, 2022.

**[SDM'23]** **S. Zeng***, J. Wang*, Z. Long, Y. Wang, H. Xiao, F. Ma, "Knowledge-Enhanced Semi-Supervised Federated Learning for Aggregating Heterogeneous Lightweight Clients in IoT", *SIAM International Conference on Data Mining*, 2023.

## Co-Authored Publications

**[NeurIPS'26]** Y. Li, Y. Feng, Z. Xu, Z. Ma, K. Zheng, F. Jiang, X. Sun, R. Shao, Z. Chen, Y. Huang, X. Han, B. Lee, K. Xu, **S. Zeng**, H. Hua, X. Zhang, B. Alomair, R. Krishna, L. Zettlemoyer, P. W. Koh, B. Ramasubramanian, L. Niu, X. Yue, R. Poovendran, "JobBench: Aligning Agent Work With Human Will", *Conference on Neural Information Processing Systems, Datasets & Benchmarks Track*, 2026. **(adopted by Meta Muse Spark 1.1, Moonshot Kimi K3, and Alibaba Qwen 3.8 Max)**

**[EMNLP'26]** K. Guo, X. Dai, **S. Zeng**, H. Shomer, H. Han, Y. Wang, J. Tang, "Beyond Static Retrieval: Opportunities and Pitfalls of Iterative Retrieval in GraphRAG", *Conference on Empirical Methods in Natural Language Processing*, 2026.

**[ICML'26]** X. Dai, K. Yang, C. Luo, **S. Zeng**, K. Guo, J. Tang, "When Do Hallucinations Arise? A Graph Perspective on the Evolution of Path Reuse and Path Compression", *International Conference on Machine Learning*, 2026.

**[KDD'26]** K. Guo, X. Dai, Z. Zhang, N. Lin, **S. Zeng**, J. Ren, H. Han, J. Tang, "Why Retrieval-Augmented Generation Fails: A Graph Perspective", *ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2026.

**[NeurIPS'25]** J. Ren, Z. Dai, X. Tang, Y. Xing, **S. Zeng**, H. Liu, J. Zeng, Q. Peng, S. Varshney, S. Wang, Q. He, C. C. Aggarwal, H. Liu, "Keeping an Eye on LLM Unlearning: The Hidden Risk and Remedy", *Conference on Neural Information Processing Systems*, 2025.

**[EMNLP'25]** K. Guo, H. Shomer, **S. Zeng**, H. Han, Y. Wang, J. Tang, "Empowering GraphRAG with Knowledge Filtering and Integration", *Conference on Empirical Methods in Natural Language Processing*, 2025.

**[ACL'25]** B. Wang, W. He, P. He, **S. Zeng**, Z. Xiang, Y. Xing, J. Tang, "Unveiling Privacy Risks in LLM agent memory", *Annual Meeting of the Association for Computational Linguistics*, 2025.

**[KDD'25]** B. Li, Z. Chen, H. Han, **S. Zeng**, J. Liu, J. Tang, "Unveiling Mode Connectivity in Graph Neural Networks", *ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2025.

**[CVPR'25]** J. Ren, K. Chen, Y. Cui, **S. Zeng**, H. Liu, Y. Xing, J. Tang, L. Lyu, "Six-CD: Benchmarking Concept Removals for Benign Text-to-Image Diffusion Models", *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.

**[TMLR]** P. He, Y. Xing, H. Xu, J. Ren, Y. Cui, **S. Zeng**, J. Tang, M. Yamada, M. Sabokrou, "Stealthy Backdoor Attack via Confidence-driven Sampling", *Transactions on Machine Learning Research*.

**[ECCV'24]** J. Ren, Y. Li, **S. Zeng**, H. Xu, L. Lyu, Y. Xing, J. Tang, "Unveiling and Mitigating Memorization in Text-to-Image Diffusion Models through Cross Attention", *European Conference on Computer Vision*, 2024.

**[EMNLP'24]** H. Xu, J. Ren, P. He, Y. Cui, **S. Zeng**, H. Liu, J. Tang, A. Liu, "On the Generalization of Training-based ChatGPT Detection Methods", *Findings of the Association for Computational Linguistics: EMNLP*, 2024.

**[ICLR'24, Spotlight]** P. He, H. Xu, J. Ren, Y. Cui, **S. Zeng**, H. Liu, C. Aggarwal, J. Tang, "Sharpness-Aware Data Poisoning Attack", *International Conference on Learning Representations, **Spotlight***, 2024.

**[NeurIPS'23]** J. Li, H. Shomer, H. Mao, **S. Zeng**, Y. Ma, N. Shah, J. Tang, D. Yin, "Evaluating Graph Neural Networks for Link Prediction: Current Pitfalls and New Benchmarking", *Conference on Neural Information Processing Systems, Datasets & Benchmarks Track*, 2023.

## Preprints and Under Review

**[Under review]** **S. Zeng**, X. Yang, T. Ma, X. Zhang, N. Wang, D. Lei, K. Sun, X. Chen, T. Stark, X. Zhu, Y. Li, R. Shao, T. Wei, R. Wanga, A. Kumar, J. Tang, W.-t. Yih, X. L. Dong, "ProPerAgent: Toward Memory-Driven Proactive Assistance", *Submitted to ICLR 2027* [arXiv].

**[Under review]** K. Guo, X. Dai, **S. Zeng**, Z. Chen, X. Long, J. Ding, J. Ren, H. Han, H. Liu, "Beyond Ranking: Making Long-Term Multi-Modal Memory Addressable", *Submitted to ICLR 2027*.

**[Under review]** X. Dai, B. Gao, C. Luo, Y. Zhang, X. Zhou, Y. Yu, **S. Zeng**, K. Guo, J. Tang, W. Feng, F. Liu, "GRAFT: Growing Missing Knowledge through Topology-Guided Data Selection", *Submitted to ICLR 2027*.

**[Preprint]** X. Long, Z. Chen, **S. Zeng**, S. Wang, K. Guo, J. Tang, "MemTrace: Probing What Final Accuracy Misses in Long-Term Memory".

**[Preprint]** X. Dai, C.-H. Lo, K. Guo, **S. Zeng**, D. Luo, J. Tang, "Uncovering Graph Reasoning in Decoder-only Transformers with Circuit Tracing".

**[Under review]** Z. Chen, J. Gu, J. Yin, X. Long, **S. Zeng**, X. Liu, K. Guo, K. Zhou, J. Tang, "Exploring Cross-Scenario Generality of Agentic Memory Systems: Diagnostics and a Strong Baseline", *ACL Rolling Review*.

**[Under review]** X. Dai, K. Guo, C.-H. Lo, **S. Zeng**, J. Ding, D. Luo, S. Mukherjee, J. Tang, "GraphGhosts: Tracing Reasoning Structures Behind Large Language Models", *ACL Rolling Review*.

**[Preprint]** Y. Lin, P. He, H. Zhao, **S. Zeng**, Z. Xiang, H. Liu, J. Tang, "Structural Alignment Faking: Hiding Malicious Capabilities in Mixture-of-Experts LLMs".

**[Preprint]** J. Ren, H. Xu, P. He, Y. Cui, **S. Zeng**, J. Zhang, H. Wen, J. Ding, H. Liu, Y. Chang, J. Tang, "Copyright Protection in Generative AI: A Technical Perspective".

## Invited Talks

- "Securing Text and Empowering Vision: Evolution from RAG to Multimodal Agentic AI" 2026
  Invited talk at University of Georgia, hosted by Prof. Zhen Xiang *(upcoming)*
- "Securing Text and Empowering Vision: Evolution from RAG to Multimodal Agentic AI" 2026
  Invited talk at Washington University in St. Louis, hosted by Prof. Jiaxin Huang
- "Attn-GS: Attention-Guided Context Compression for Efficient Personalized LLMs" 2026
  Oral presentation at ACL 2026
- "Towards Knowledge Checking in Retrieval-Augmented Generation: A Representation Perspective" 2025
  Oral presentation at NAACL 2025
- "Heterogeneous Federated Learning via Grouped Sequential-to-Parallel Training" 2022
  Oral presentation at DASFAA 2022

## Open Source Contributions

- **Founding Contributor:** SimpleJev, open infrastructure for System 1 thinking models: single-pass fast decision-making, with open models, training framework, and evaluation benchmarks.
- **Contributor:** OpenRSI Index, open evaluation of recursive self-improvement in AI agents: real-world research tasks packaged as agent environments, scored by improvement over human baselines.

## Community Services

- **Workshop Organizer:** The 2nd Workshop on Efficient Reasoning [COLM'26].
- **Conference Reviewer / Program Committee:** NeurIPS, ICML, ICLR; ACL, EMNLP, NAACL (via ACL Rolling Review), etc.
- **Journal Reviewer:** ACM Transactions on Information Systems (TOIS), IEEE TKDE, IEEE TMC, IEEE TPDS, IEEE Transactions on Information Forensics & Security, IEEE Transactions on Services Computing, IEEE Transactions on Vehicular Technology, IEEE Internet of Things Journal, etc.
- Reviewed **60**+ papers.

## Mentorship

- **Jiankun Zhang** (PhD@Jilin University): [ICML'26], [ACL'24]
- **Xianxuan Long** (PhD@MSU): [NeurIPS'26], [Preprint]
- **Bo Wang** (PhD@Jilin University): [ACL'25]
- **Qirui Wang** (MS@Xi'an Jiaotong University): [NeurIPS'26]